Relevant Literature & Details

▼ ContractEval

1. Single-Agent Architecture Only

The Weakness

ContractEval evaluates each model in isolation—one model, one contract, one question. There's no assessment of:

- Agent coordination overhead
- Specialized agent performance vs. generalist
- Information flow between processing stages
- Error propagation across components

Your Thesis Opportunity

Contribution: First empirical comparison of multi-agent vs. single-agent architectures on CUAD-based contract tasks.

Concrete extension:

```
ContractEval baseline:    Single LLM → Contract + Question → Extracted Clause
Your evaluation:          Orchestrator → [Risk Agent, Clause Agent, ...] → Aggregated Output
```

Metrics to add:

New Metric	What It Measures
Coordination overhead	Time cost of inter-agent communication
Agent agreement rate	How often specialized agents concur on findings
Error containment	Whether failures cascade or stay isolated
Ablation delta	Performance drop when removing each agent

Research question this addresses: RQ2 (multi-agent vs. single-agent comparison)

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2. Single-Contract Isolation

The Weakness

ContractEval treats each contract as independent. Real CLM involves:

- Master agreements with amendments
- Back-to-back contracts that must align
- Portfolio-level risk assessment
- Clause consistency across related deals

The paper never tests whether a model can identify **inconsistencies between contracts** or maintain **consistency across a contract portfolio**.

Your Thesis Opportunity

Contribution: Cross-contract consistency evaluation framework.

New task formulation:

```
Input:  Contract A (Master Agreement) + Contract B (Subcontract)
Query:  "Do liability caps in Contract B align with Contract A?"
Output: {
    "aligned": false,
    "contract_a_clause": "Liability capped at $1M...",
    "contract_b_clause": "Unlimited liability for...",
    "conflict_explanation": "Subcontract exposes greater liability than master permits"
}
```

This directly validates your Clause Alignment Agent design.

Evaluation metrics:

- Cross-contract consistency detection accuracy
- False alarm rate (flagging non-issues)
- Conflict severity ranking correlation with expert judgment

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3. No Explainability Assessment

The Weakness

ContractEval measures **what** models extract, not **why** they extracted it. The paper acknowledges this gap but doesn't address it:

> "how can evaluation methods for contract review and legal tasks be better aligned with the standards and expectations of senior legal professionals?"

For enterprise CLM, explainability isn't optional—it's required for:

- GDPR Article 22 (right to explanation)
- EU AI Act transparency requirements
- Internal audit and compliance
- Building user trust

Your Thesis Opportunity

Contribution: Explainability metrics for contract AI systems.

Dimensions to evaluate:

Dimension	Definition	Measurement Approach
Faithfulness	Does explanation reflect actual reasoning?	Perturbation tests—modify input, check if explanation changes appropriately
Completeness	Are all relevant factors mentioned?	Coverage of legal elements identified by experts
Comprehensibility	Can a lawyer understand it?	Human evaluation (Likert scale)
Actionability	Does explanation support decision-making?	Task-based evaluation—can user act on explanation?

Operationalization via LangFuse:

python

```
# Your system captures reasoning traces
{
    "extraction": "Liability capped at $500,000...",
    "reasoning_trace": [
        "Step 1: Identified 'limitation of liability' section header",
        "Step 2: Located monetary cap language ('shall not exceed')",
        "Step 3: Extracted specific amount and scope",
        "Step 4: Verified no exceptions apply to this cap"
    ],
    "confidence": 0.89,
    "relevant_context_spans": ["Section 8.1", "Section 8.3(a)"]
}
```

```
**Research question this addresses:** RQ3 (explainability mechanisms)

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## 4. Binary Span Matching (No Partial Credit)

### The Weakness
ContractEval's TP definition requires the prediction to "fully cover the labeled span."
This creates problematic edge cases:

**Example:**
```
Ground truth: "This Agreement shall automatically renew for successive
               one (1) year periods, unless either party provides written
               notice of non-renewal at least ninety (90) days prior."

Prediction: "automatically renew for successive one (1) year periods"

ContractEval result: FN (doesn't fully cover)
Reality: This is a useful partial extraction
```
```

This binary approach:

- Penalizes reasonable partial extractions
- Doesn't distinguish "almost right" from "completely wrong"
- May underestimate practical utility

Your Thesis Opportunity

Contribution: Graduated evaluation framework with partial credit.

Adopt SemEval-2013 Task 9 span evaluation:

Match Type	Definition	Score Weight
Strict	Exact boundary AND correct category	1.0
Exact	Correct boundaries, any category	0.9
Partial	Overlapping span, correct category	0.5
Type	Any overlap, correct category	0.25

Extended metrics:

python

```
def graduated_score(prediction, ground_truth):
    overlap = compute_overlap(prediction, ground_truth)

    if overlap == 1.0 and boundaries_exact(prediction, ground_truth):
        return 1.0  # Strict match
    elif overlap >= 0.8:
```

```
        return 0.75  # Near-complete extraction
    elif overlap >= 0.5:
        return 0.5   # Partial extraction
    elif overlap > 0:
        return 0.25  # Minimal overlap
    else:
        return 0.0   # Complete miss
```

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5. No Statistical Rigor

The Weakness
ContractEval reports point estimates without:
- Confidence intervals
- Statistical significance tests
- Multiple runs with different seeds
- Prompt sensitivity analysis

This violates ACL reproducibility standards (Dror et al., 2018) and limits reliability of comparative claims.

Example problematic claim:
> "Qwen3 8B (thinking) achieves F1 of 0.540 vs. GPT-4.1's 0.641"

Without confidence intervals, we don't know if this 10% gap is statistically meaningful or within noise.

Your Thesis Opportunity

Contribution: Statistically rigorous evaluation methodology for legal AI.

Implementation:
1. **Multiple seeds:** Run each model 5+ times with different random seeds
2. **Bootstrap confidence intervals:** 1,000+ samples for stable estimates
3. **Significance testing:** McNemar's test for paired comparisons
4. **Effect size reporting:** Cohen's d alongside p-values
5. **Prompt sensitivity:** Test 5+ prompt variants per model

Reporting template:
```
Risk Analysis Agent Performance:
- F1: 0.72 (95% CI: 0.69-0.75)
- Improvement over single-agent baseline: +8.3% (p < 0.01, Cohen's d = 0.65)
- Prompt sensitivity: σ = 0.03 across 5 prompt variants
```

This directly addresses your supervisor's emphasis on "robust evaluation frameworks" and "statistical rigor."

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6. Static Extraction Only (No Lifecycle Coverage)

The Weakness

```
ContractEval only tests the **review** phase of contract lifecycle:
```
```
ContractEval coverage:
[Drafting] → [Negotiation] → [Review ✓] → [Execution] → [Obligation Tracking] → [Renewal]
```

Real CLM systems must handle:

- **Pre-signature:** Drafting, redlining, version comparison
- **Post-signature:** Obligation monitoring, deadline tracking, renewal management

Your Thesis Opportunity

Contribution: Extended CLM task taxonomy with lifecycle-phase-specific evaluation.

New task definitions:

Phase	Task	Input	Expected Output
Review	Clause extraction (ContractEval)	Contract + Question	Extracted span
Review	Risk scoring	Contract	Risk scores per clause type
Post-signature	Obligation extraction	Contract	List of obligations with parties, deadlines
Post-signature	Deadline monitoring	Contract + Current date	Upcoming deadlines, overdue items
Cross-contract	Consistency checking	Contract set	Inconsistencies, conflicts

This validates your Obligation Tracking Agent and extends evaluation beyond ContractEval's scope.

Sample obligation extraction evaluation:

python

```
# Ground truth obligation
{
    "obligation": "Licensee shall pay royalty within 30 days of quarter end",
    "obligor": "Licensee",
    "obligee": "Licensor",
    "deadline_type": "recurring",
    "deadline_rule": "30 days after quarter end",
    "next_deadline": "2026-01-30"
}
```

```
# Metrics
- Obligation detection recall (did we find all obligations?)
- Entity extraction accuracy (correct obligor/obligee?)
- Deadline parsing accuracy (correct date computation?)
```

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7. No Reasoning Quality Assessment

The Weakness
ContractEval can't distinguish between:

Model A: Extracts correct clause through pattern matching (fragile)
```
Input: "What is the governing law?"
Process: Regex match on "governed by the laws of [STATE]"
Output: "governed by the laws of Delaware"
```

Model B: Extracts correct clause through legal reasoning (robust)
```
Input: "What is the governing law?"
Process:
  1. Identify choice-of-law section
  2. Parse governing law clause structure
  3. Check for exceptions or carve-outs
  4. Verify no conflicting provisions elsewhere
Output: "governed by the laws of Delaware"
```

Both get the same F1 score, but Model B is more reliable for edge cases.

Your Thesis Opportunity

Contribution: Reasoning quality metrics for legal AI.

Evaluation dimensions:

Dimension	Question	Measurement
Reasoning depth	How many logical steps?	Count steps in CoT trace
Legal grounding	Are legal concepts correctly applied?	Expert annotation of reasoning traces
Consistency	Same reasoning for similar cases?	Variance across semantically similar queries
Robustness	Does reasoning hold under perturbation?	Adversarial input testing

Layered CoT evaluation (from your literature review):

python

```
reasoning_trace = [
    {"step": 1, "action": "locate_section", "confidence": 0.95},
    {"step": 2, "action": "parse_clause", "confidence": 0.88},
    {"step": 3, "action": "check_exceptions", "confidence": 0.92},
    {"step": 4, "action": "verify_consistency", "confidence": 0.85}
]

# Metrics
trace_completeness = len(reasoning_trace) / expected_steps
avg_confidence = mean([s["confidence"] for s in reasoning_trace])
verification_coverage = count_verification_steps / total_steps
```

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8. Category Imbalance Not Addressed

The Weakness
ContractEval documents that both GPT-4.1 and open-source models achieve near-zero F1 on rare categories (Uncapped Liability, Joint IP Ownership), but offers no solution.

The paper essentially says: "This is a problem" without proposing remediation.

Your Thesis Opportunity

Contribution: Specialized agents for underperforming clause categories.

Hypothesis: A dedicated agent trained/prompted specifically for rare, high-risk clauses will outperform a generalist model.

Experimental design:
```
Baseline: Single generalist model on all 41 categories

Treatment 1: Generalist + Specialist agent for liability clauses
Treatment 2: Generalist + Specialist agents for top-5 failure categories
Treatment 3: Full multi-agent with category-specialized routing
```

Analysis:
- Does specialization improve rare category performance?
- What's the trade-off with common category performance?
- Is there an optimal specialization granularity?

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9. No Practical Deployment Metrics

The Weakness
ContractEval reports F1, F2, and Jaccard, but ignores:
- **Latency:** How long does extraction take?
- **Cost:** What's the API cost per contract?
- **Throughput:** Contracts processed per hour
- **Resource utilization:** GPU/memory requirements

For enterprise deployment, a system with 5% lower F1 but 10x lower cost might be

```
preferable.

### Your Thesis Opportunity

**Contribution:** Cost-performance Pareto analysis for CLM systems.

**Extended metrics table:**

| System | F1 | F2 | Latency (s) | Cost/Contract | Throughput |
|--------|----|----|-------------|---------------|------------|
| GPT-4.1 (baseline) | 0.64 | 0.68 | 45 | $0.50 | 80/hr |
| Your multi-agent | 0.72 | 0.75 | 120 | $1.20 | 30/hr |
| Your multi-agent (optimized) | 0.70 | 0.73 | 60 | $0.75 | 60/hr |

**Pareto frontier visualization** showing accuracy vs. cost trade-offs would be
valuable for practitioners.

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## 10. No Iterative Refinement

### The Weakness
ContractEval is single-shot: one query, one response, done.

Real contract review is iterative:
1. Initial extraction
2. Lawyer reviews, asks clarifying questions
3. Model refines extraction
4. Repeat until satisfactory

### Your Thesis Opportunity

**Contribution:** Multi-turn evaluation protocol for contract AI.

**Task extension:**
```
Turn 1:
  Query: "What are the termination provisions?"
  Response: [Initial extraction]

Turn 2:
  Feedback: "You missed the termination-for-cause clause in Section 8.2"
  Response: [Refined extraction including Section 8.2]

Turn 3:
  Feedback: "Is there a cure period before termination for cause?"
  Response: [Additional extraction with cure period details]
```
```

Metrics:

- Refinement responsiveness (does model improve with feedback?)
- Feedback incorporation accuracy
- Convergence speed (turns to reach satisfactory extraction)

Summary: Mapping Weaknesses to Your Contributions

ContractEval Weakness	Your Thesis Contribution	Relevant RQ
Single-agent only	Multi-agent vs. single-agent empirical comparison	RQ2
Single-contract isolation	Cross-contract consistency evaluation	RQ1 (Clause Alignment Agent)
No explainability	Explainability metrics + LangFuse integration	RQ3
Binary span matching	Graduated partial-credit evaluation	Methodology
No statistical rigor	Bootstrap CIs, significance tests, effect sizes	Methodology
Review phase only	Obligation tracking evaluation	RQ1 (Obligation Tracking Agent)
No reasoning assessment	Layered CoT quality metrics	RQ3
Category imbalance	Specialized agents for rare clauses	RQ1
No deployment metrics	Cost-performance Pareto analysis	Practical contribution
No iteration	Multi-turn refinement evaluation	Extension

Recommended Prioritization for Your Thesis

Must address (core contributions):

1. Multi-agent vs. single-agent comparison (directly answers RQ2)
2. Explainability metrics (directly answers RQ3)
3. Statistical rigor (satisfies supervisor requirements)

Should address (strengthens contribution):

4. Cross-contract consistency (validates Clause Alignment Agent)
5. Obligation tracking evaluation (validates Obligation Tracking Agent)
6. Graduated evaluation metrics (improves over ContractEval methodology)

Could address (if time permits):

7. Cost-performance analysis (practical value)
8. Specialized agents for rare categories (extends RQ1)
9. Multi-turn refinement (future work section)

▼ Automated construction contract analysis for risk and responsibility assessment using natural language processing and machine learning

Irem Dikmen

https://www.sciencedirect.com/science/article/pii/S0166361525000168

▼ Applying Large Language Models for Enhancing Contract Drafting

https://ceur-ws.org/Vol-3423/paper7.pdf

▼ Proceedings of the NLLP Workshop 2025

https://aclanthology.org/2025.nllp-1.pdf